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SOFTWARE DEVELOPMENT ENVIRONMENTS

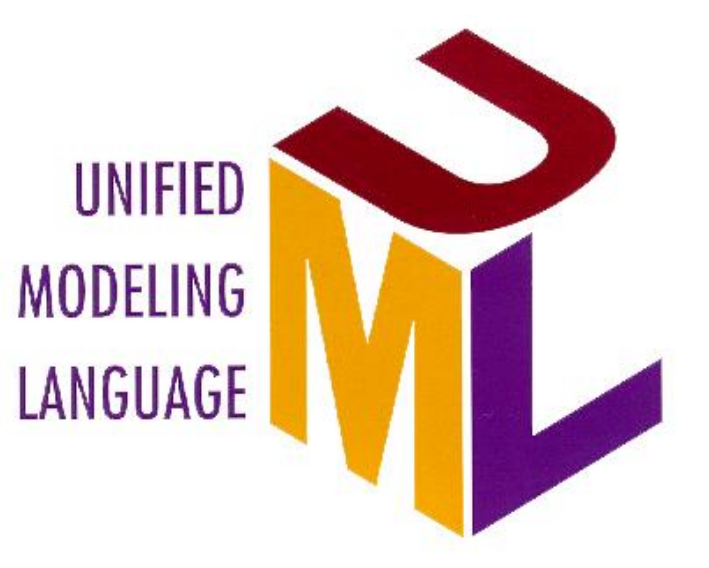
Chapter 02

Unified Modeling Language - UML

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Lesson 01

Fundamentals



Contents

1. Lesson's Objectives
2. Benefits of Visual Modeling
3. History of the UML
4. Visual Modeling with UML
5. Summary



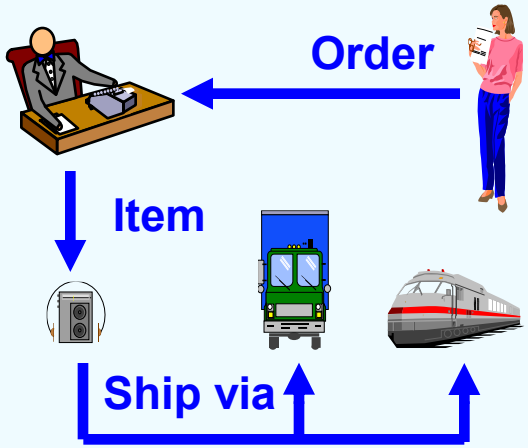
Objectives

After the lesson, you will be able to:

- Explain the benefits of visual modeling
- Explain the history and major characteristics of UML
- List types of diagrams in UML
- Explain the major usage of each diagram
- Explain the meaning of each component in every type of diagram in UML
- Model a system using UML

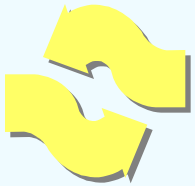


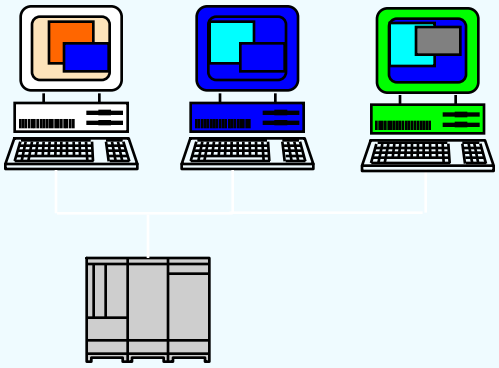
What is Visual Modeling?



Business Process

"Modeling captures essential parts of the system."
Dr. James Rumbaugh



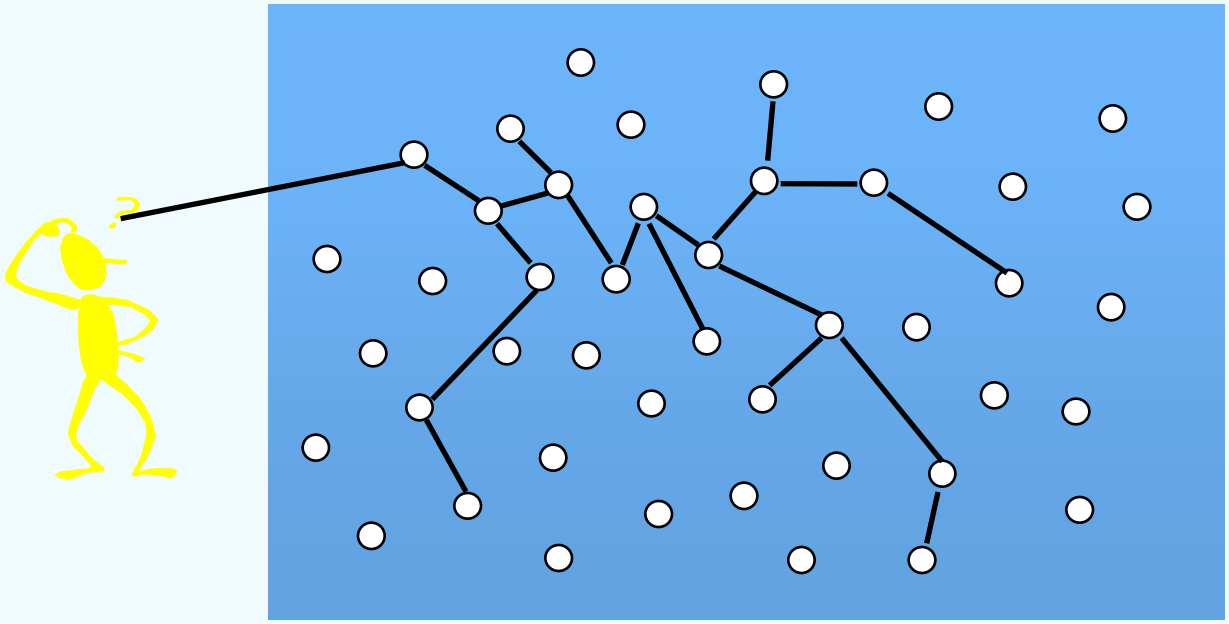


Computer System

Visual Modeling is modeling using standard graphical notations

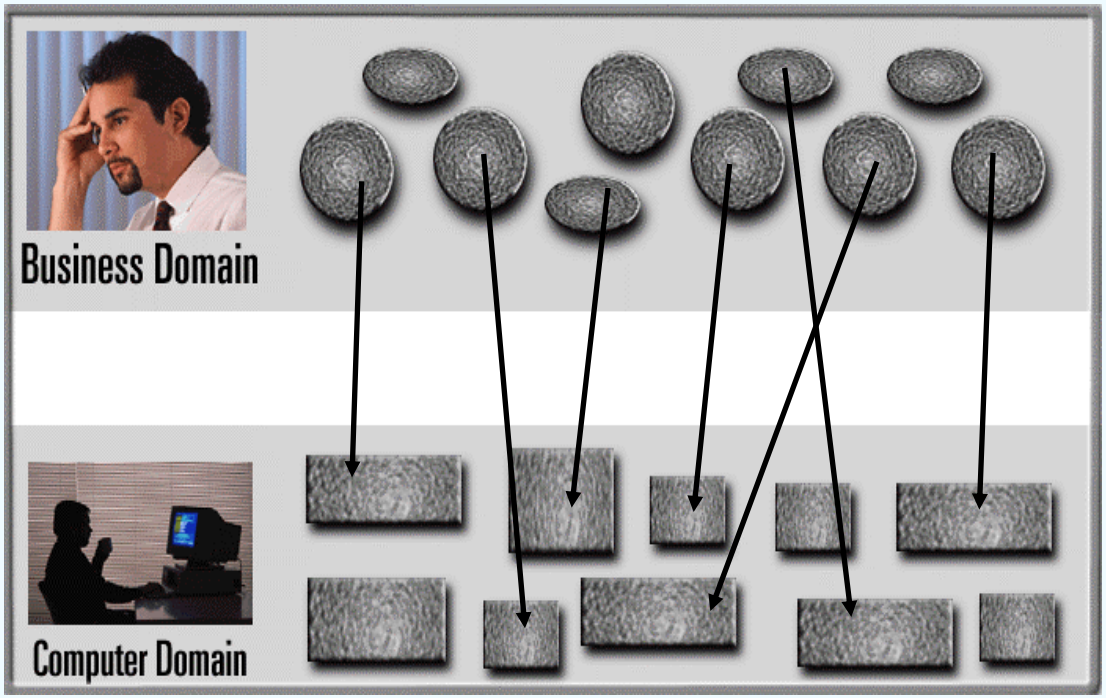
Visual Modeling Captures Business Process

Use Case Analysis is a technique to capture business process from user's perspective



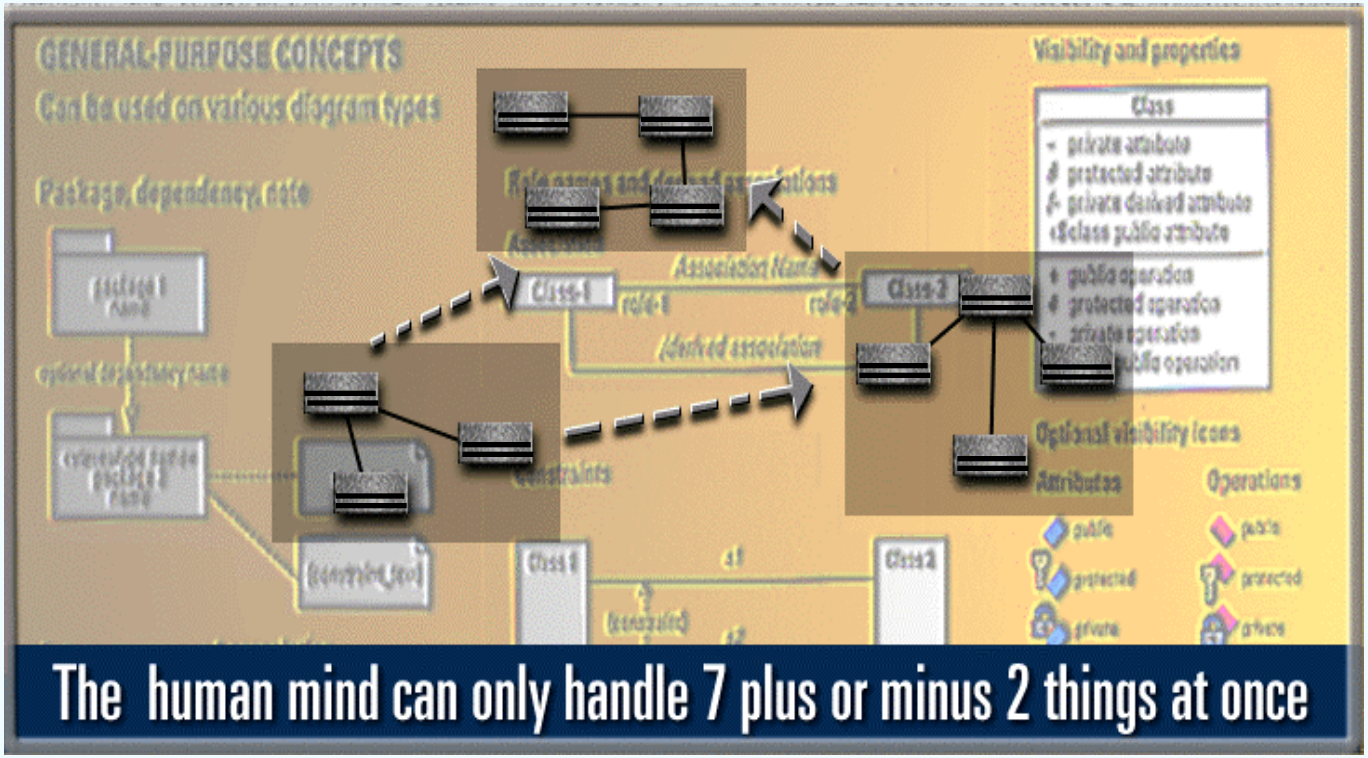
Visual Modeling is a Communication Tool

Use visual modeling to capture business objects and logic

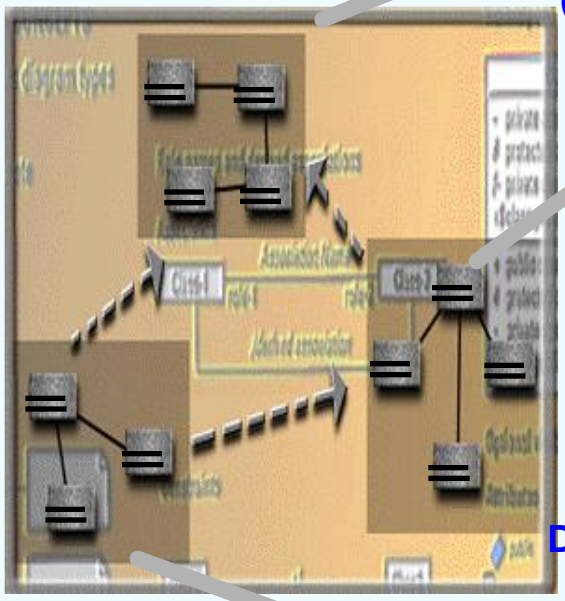


Use visual modeling to analyze and design your application

Visual Modeling Manages Complexity



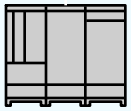
Visual Modeling Defines Software Architecture



User Interface
(Visual Basic,
Java)

Business Logic
(C++, Java)

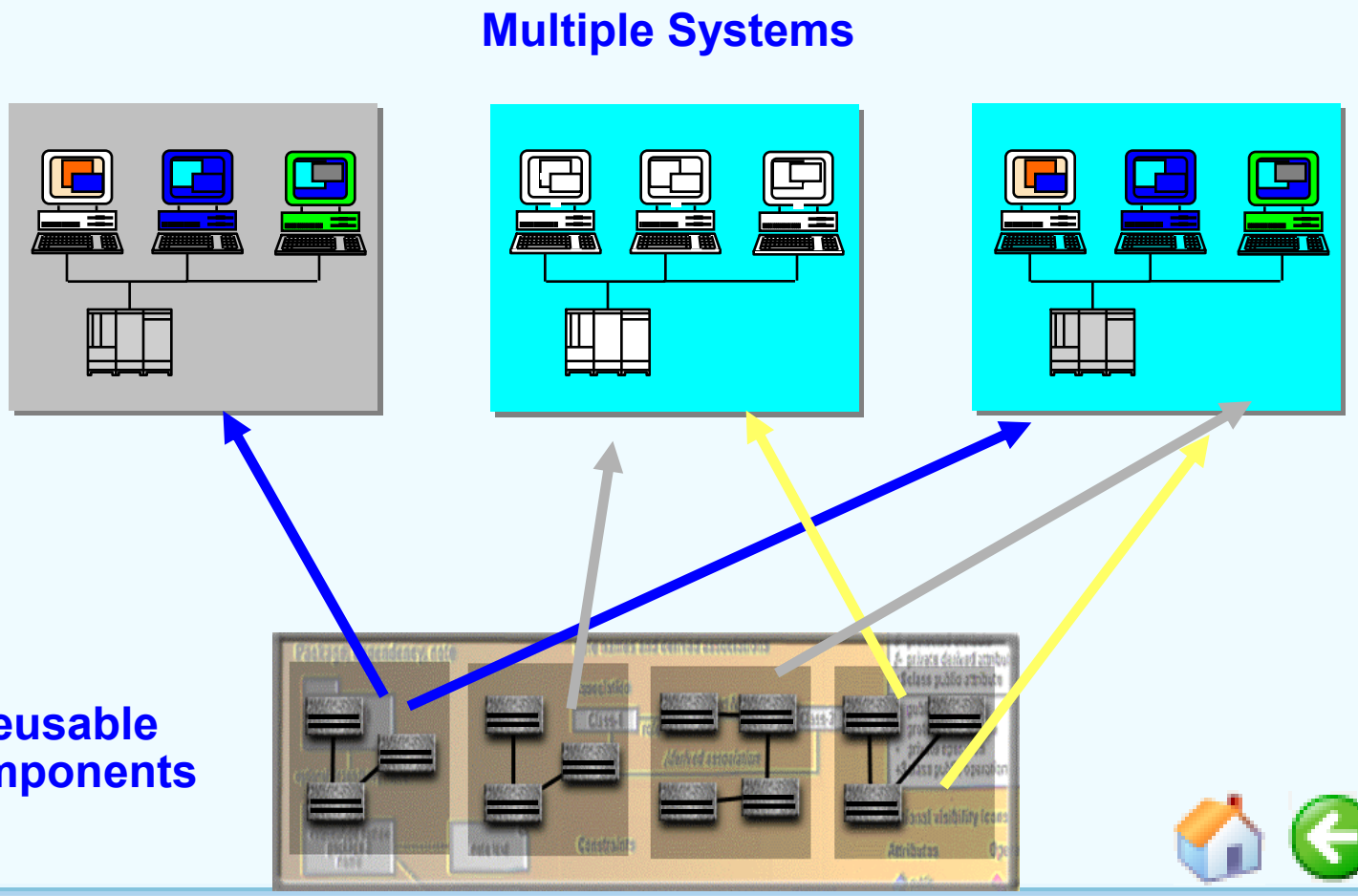




Database Server
(C++ & SQL)

Model your system independent of implementation language

Visual Modeling Promotes Reuse



What is the UML?

- **UML stands for Unified Modeling Language**
- **The UML combines the best of the best from:**
  - Data Modeling concepts (Entity Relationship Diagrams)
  - Business Modeling (work flow)
  - Object Modeling
  - Component Modeling



What is the UML? (2)

- The UML is the standard language for visualizing, specifying, constructing, and documenting the artifacts of a software-intensive system
- It can be used with all processes, throughout the development life cycle, and across different implementation technologies



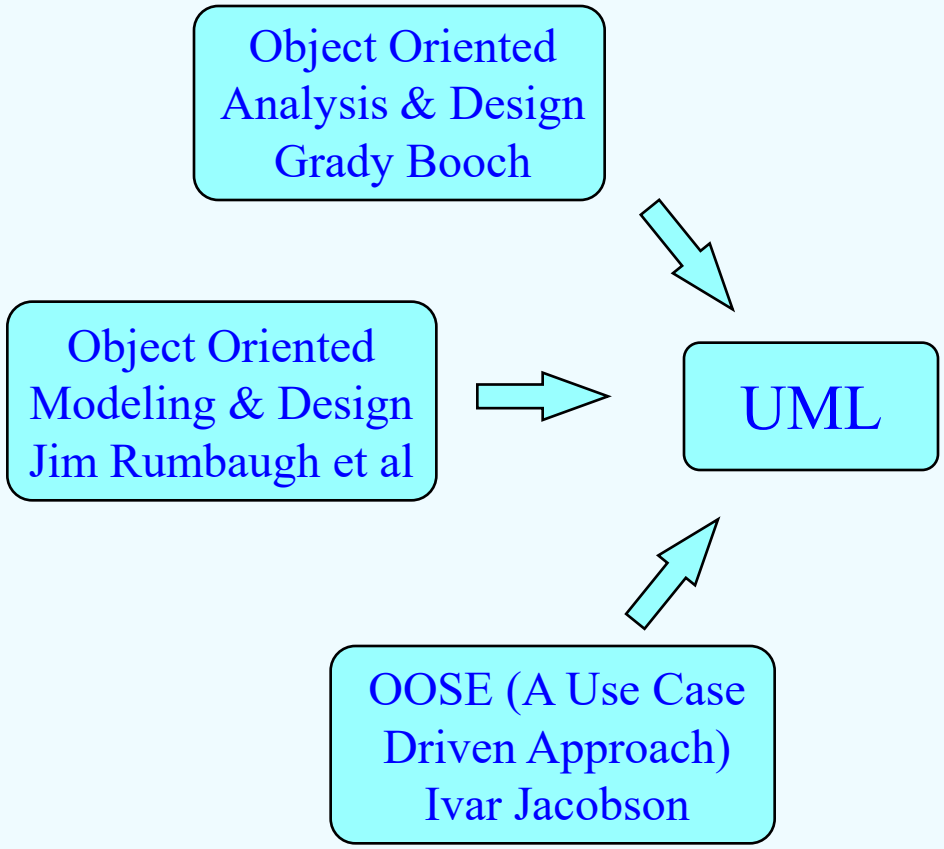
Before the UML

- 1960's - 70's
  - COBOL, FORTRAN, C
  - Structured analysis and design techniques
- 1980's - early 1990's
  - Smalltalk, Ada, C++, Visual Basic
  - Early generation OO methods
- Mid/late 1990's
  - Java
  - UML
  - Unified Process

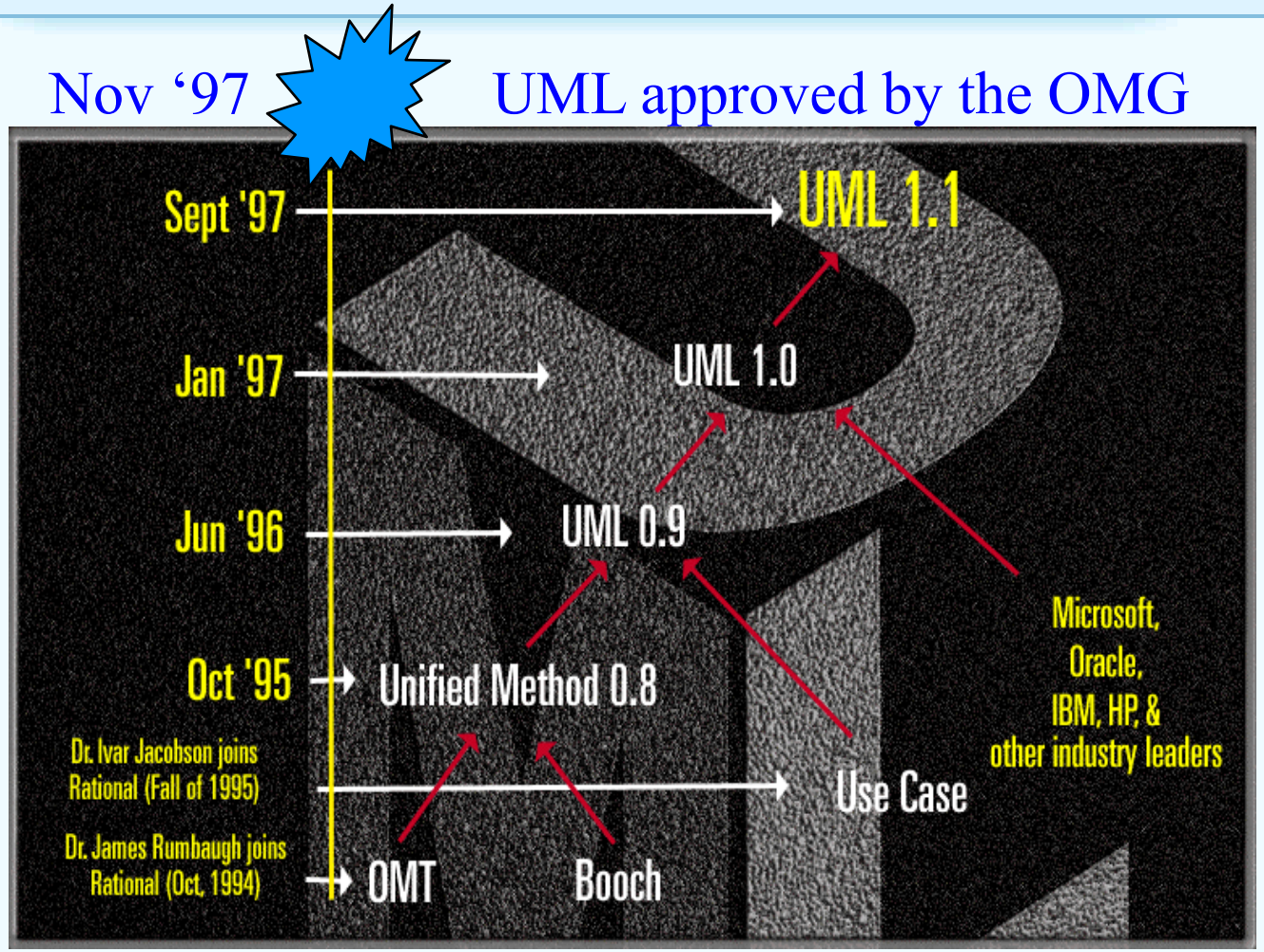


UML characteristics

- Syntax only
- Comprehensive
- Language Independent
- Process Independent
- Tool Independent



History of the UML



History of the UML (2)

- 2001: UML 1.3 approved
- 2005: UML 2.0
- 2007: UML 2.1
- 2009: UML 2.2
- 2010: UML 2.3



UML Supports Application Development

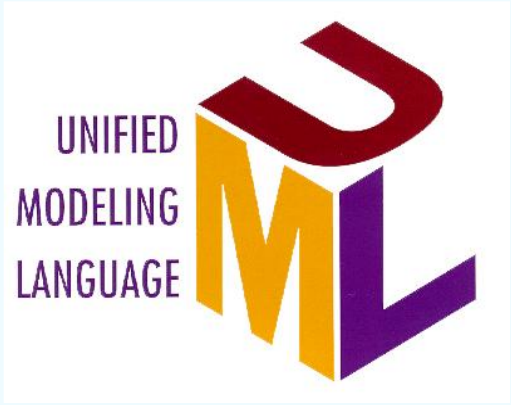
Objects

Business Objects

Relationships

large scale system

ORDBMS
Oracle



Classes

application partitioning

Components
Microsoft

Scenarios

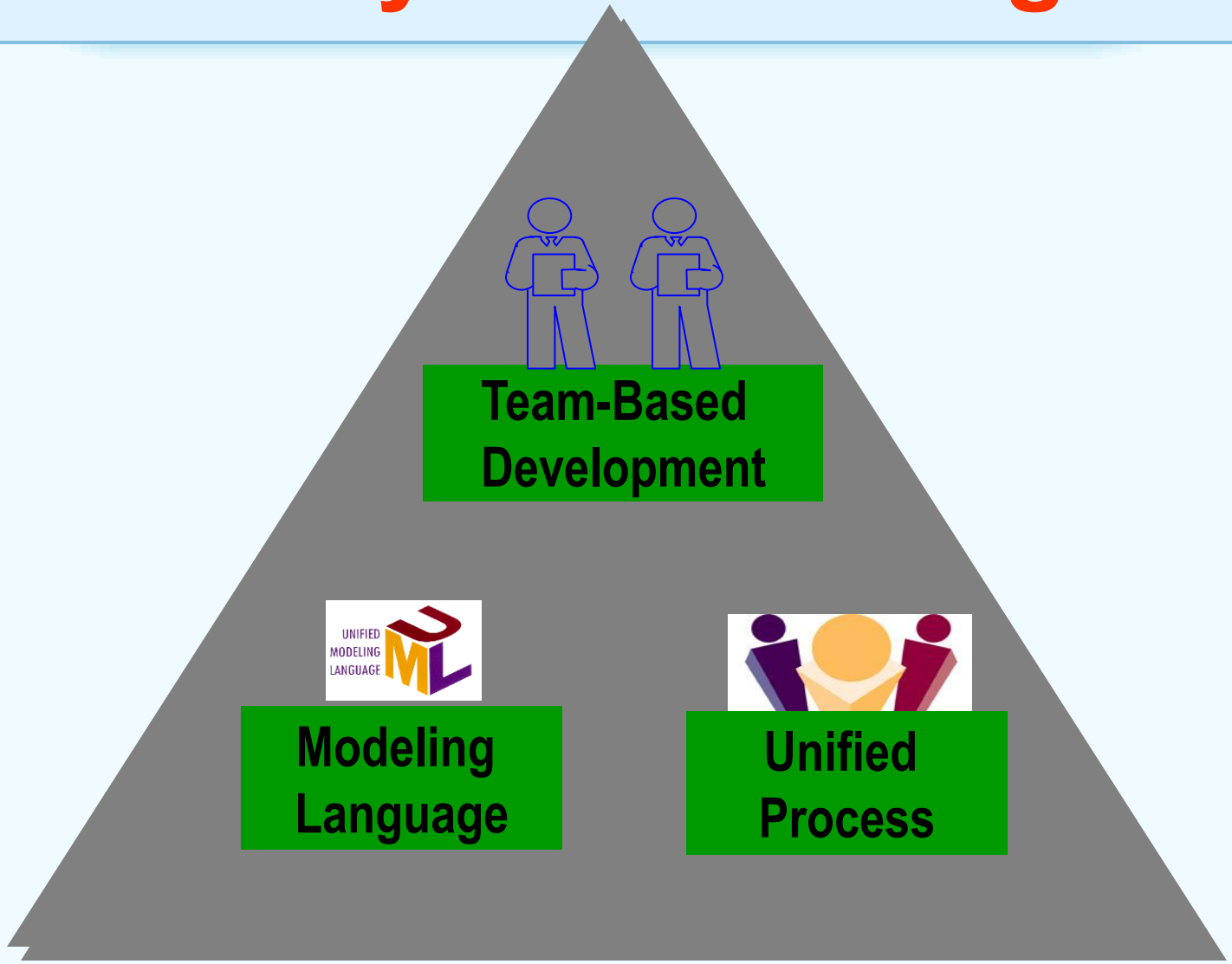
CORBA
OMG

Use Cases

ActiveX/COM
Microsoft

Business Process

UML only is not enough!



UML's fields of application

- **The UML may be used to:**
  - Display the boundary of a system & its major functions using use cases and actors
  - Illustrate use case realizations with interaction diagrams
  - Represent a static structure of a system using class diagrams



UML's fields of application (2)

- **The UML may be used to:**
  - Model the behavior of objects with state transition diagrams
  - Reveal the physical implementation architecture with component & deployment diagrams
  - Extend your functionality with stereotypes

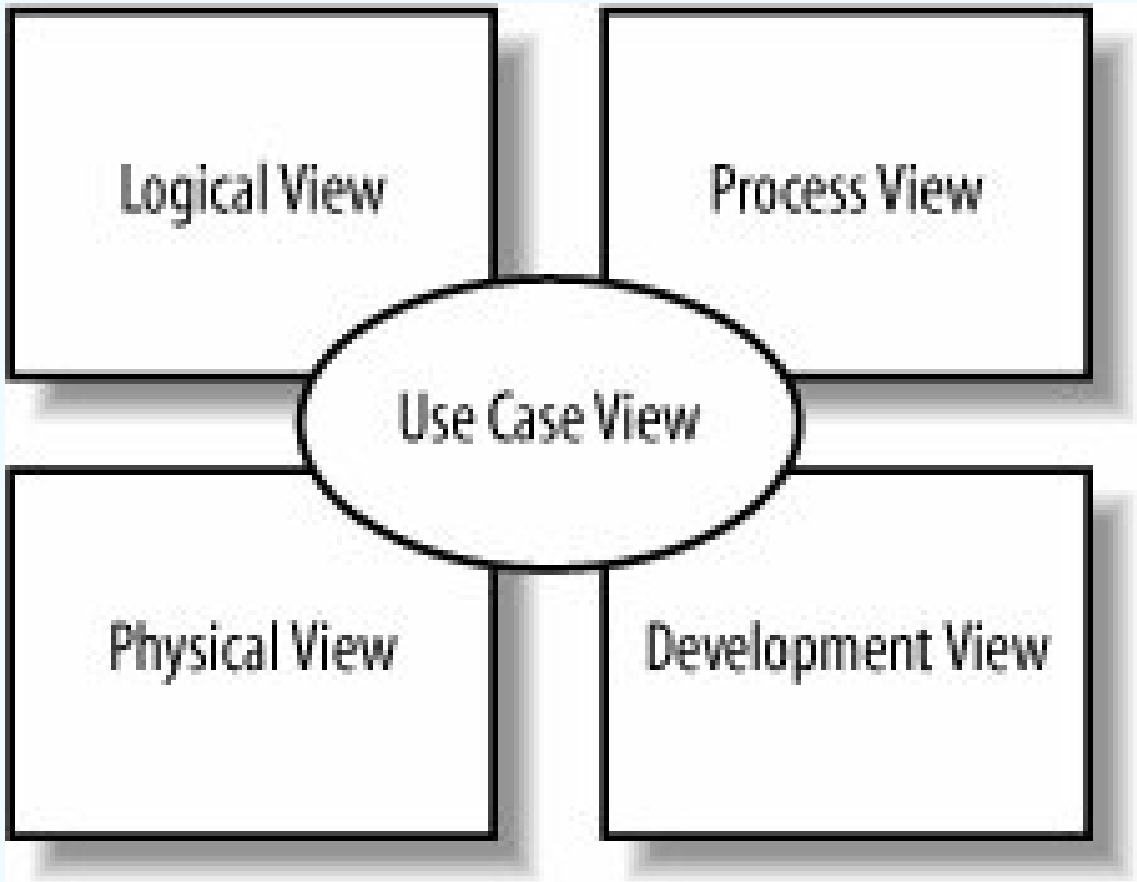


Visual modeling with UML

- **Contents:**
  - The 4+1 view
  - Use case view
  - Logical view
  - Process view
  - Component view
  - Development view



The 4+1 view

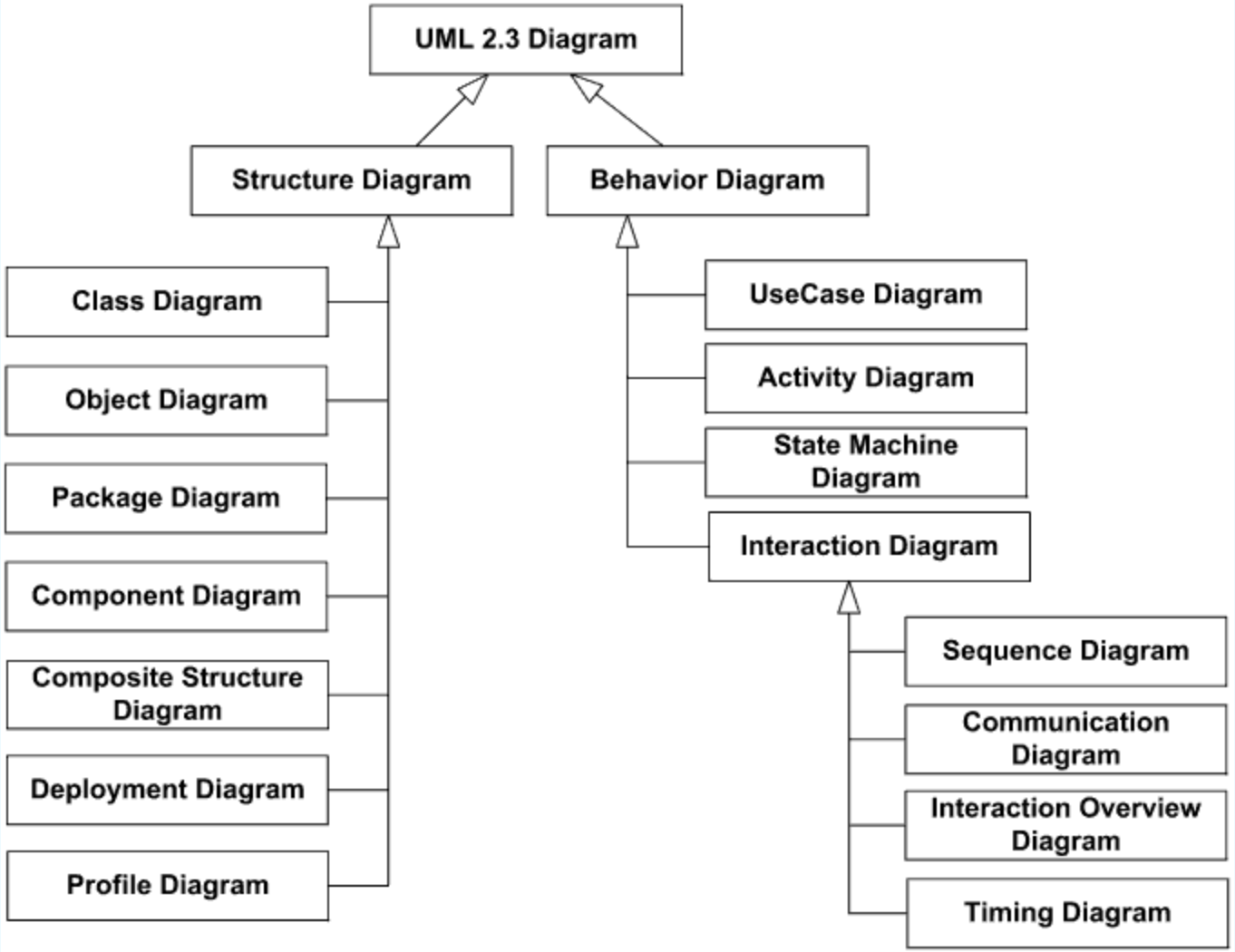

Logical View
Process View
Use Case View
Physical View
Development View

UML diagram types

- **Use Case Diagrams**
- **Static Structure Diagrams**
  - Object Diagrams
  - Class Diagrams
- **Interaction Diagrams**
  - Sequence Diagrams
  - Collaboration Diagrams
- **Statechart Diagrams**
- **Activity Diagrams**
- **Implementation Diagrams**
  - Component Diagrams
  - Deployment Diagrams



Models and Diagrams



Use Case View Diagrams

- **Viewed in one or more diagrams**
- **Contains any or all of the following diagrams**
  - Use case diagrams: packages, actors, use cases, and relationships
  - Interaction diagrams (sequence diagrams and/or collaboration diagrams): objects and messages



Use Case Diagram

- **A use case diagram shows what should be performed by the system and what exists outside the system**
  - It shows the existence of actors (outside the system), use cases (what should be performed by the system), and their relationships
  - A use-case diagram may depict all or some of the use cases of a system



Interaction Diagrams

- Interaction diagrams show the existence of objects and their relationships in the logical design of a system
- They may represent all or part of the object structure of a system



Interaction Diagrams (2)

- **There are two main kinds of interaction diagrams, each shows a different view of the same interaction**
  - A sequence diagram shows object interactions arranged in time sequence
  - A collaboration diagram is organized around the objects and their links to each other



Logical View Diagrams

- **Elements in the logical view may be viewed in one or more diagrams**
- **The logical view contains any or all of the following diagrams**
  - Class diagrams
    - Packages, classes, and relationships
  - State diagrams
    - States, events, and transitions



Class Diagram

- **A class diagram depicts the static view of the system**
  - It shows the existence of packages, classes, and their relationships in the logical design of a system
  - A class diagram may represent all or part of the class structure of a system
- **There are typically many class diagrams in a model**



State Diagram

- **A state diagram is created for classes with significant dynamic behavior**
- **A state diagram shows:**
  - The state space of a given class
  - The events that cause a transition
  - The actions that result from a state change



Component View Diagrams

- Elements in the component view may be viewed in one or more component diagrams
- A component diagram shows the allocation of classes and objects to components in the physical design of a system
  - A component diagram may represent all or part of the component architecture of a system
- A component diagram shows packages, components, and dependencies



Deployment View Diagrams

- Elements in the deployment view may be viewed in the deployment diagram
  - The deployment view contains only one deployment diagram
- The deployment diagram shows the allocation of processes to processors in the physical design of a system
- A process diagram shows
  - Nodes
  - Connections



Summary

- Visual modeling make it easier to describe the real world
- UML is used in various fields of application
- The latest version of UML is 2.3
- A model can be view in different aspects (4+1 view)
- UML uses kinds of diagrams to model the real world. Each diagram has its own components and functions





THE END

